Indo-Quantum Summit Gujarat

Peaked Quantum Circuits and Classical Inferability of Dominant Bitstrings

Shriyansh Aggarwal
24ucs031@lnmiit.ac.in

Pandit Deendayal Energy University, Gandhinagar, Gujarat, India

Motivation

Quantum advantage via sampling faces a **verification challenge**, i.e., it is non-trivial to classically determine whether the output of a circuit is correct. However, one such proposed property of the output to weakly verify the evidence of correct behavior is the presence of a **dominant bitstring.**

If a circuit's distribution is sharply concentrated, the peak can serve as a concise classical certificate. These circuits are known as **peaked circuits**. Recent results show that generic random quantum circuits almost never produce such peaks, implying that any dominant bitstring must arise from added circuit structure.

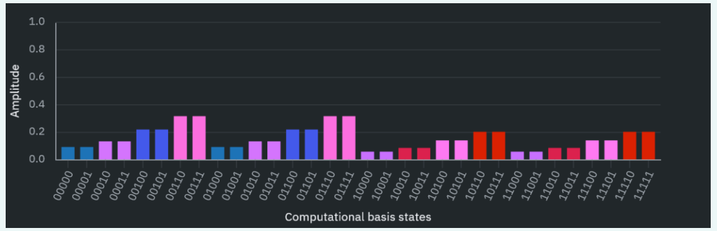


Fig 1. Random Quantum Circuit

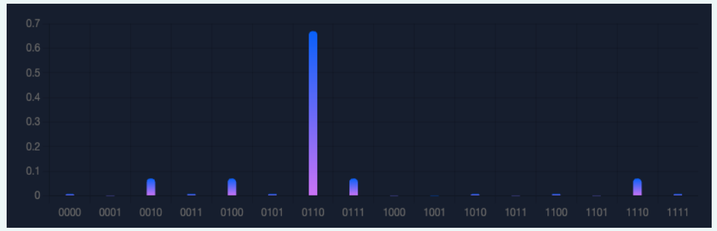


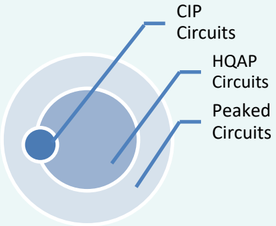
Fig 2. Peaked Quantum Circuit

Methodology

Circuit Definition → Structural Analysis → Inferability Assessment

The focus of this study is to identify and categorize **embedded substructure** in peaked quantum circuits. Previous works show that many such circuits are classically intractable to simulate on consumer-grade hardware, claiming potential cryptographic advantages linked to the "**quantum supremacy**" goal.

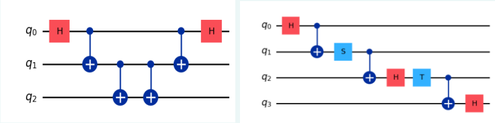
Q. Does the structure that creates dominant bitstrings also make it classically inferable from the circuit description?



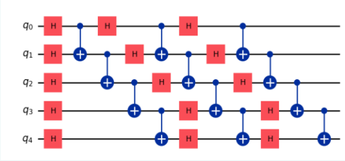
Peaked circuit families

CIP: Classically Inferable Peaked Circuits

HQAP: Heuristic Quantum Advantage Peaked Circuits



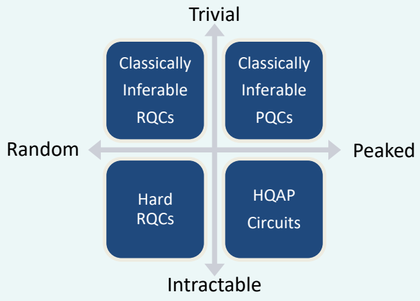
Embedded classical behavior or symmetry results in inferable peaks.



Departure from randomness: Peak inducing quantum circuits that contain **repeated patterns**, structure that fail to hide the dominant state fall under **CIP** circuits.

Note: These circuit examples only serve to demonstrate the structural patterns observed in many common peaked circuits.

Results



1. Peakedness of a quantum state is **insufficient** for verification.
2. Existing literature addresses the existence of a peak, but its **inferability** has not been studied yet.
3. PQCs can be **classified** based on ease of classical verification.

Conclusion & Further Work

This work studies the **two classes** of peaked quantum circuits to clarify previous related papers further into verifiable quantum advantage. Papers like **Aaronson et al. (2024)** and **Gharibyan et al. (2025)** establish the foundations of peaked behavior arising from random circuits containing hidden structure and construction of **HQAP** circuits.

Future work aims to **formalize distinguishing criteria** for inferable and non-inferable peaked circuits, to better understand which structural features expose or **conceal dominant outputs**, and to assess the implications of this distinction for verification and benchmarking protocols in the **NISQ era**.

References & Links

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